

Sentinel-IoT: An Autonomous Explainable XDR Framework for Heterogeneous Industrial IoT Security and Compliance Auditing

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Abstract

The proliferation of heterogeneous Industrial Internet of Things (IIoT) ecosystems has exacerbated vulnerability to sophisticated, multi-vector cyber-attacks, including distributed denial-of-service, process injection, and ransomware exfiltration. Traditional intrusion detection systems (IDS) suffer from fundamental architectural bottlenecks: sequential recurrent models (e.g., CNN-BiLSTM) incur substantial GPU latency scaling at $\mathcal{O}(T)$, operate as opaque black-box classifiers without verifiable mathematical proofs, stop short of automated threat remediation, and remain divorced from industrial regulatory compliance auditing. To overcome these limitations, this paper presents **Sentinel-IoT**, an autonomous, cross-layer Extended Detection and Response (XDR) framework. Sentinel-IoT features a dual-head **Google Conformer** core that interleaves Depthwise Separable Convolutions with Multi-Head Self-Attention in a parallel Macaron-style sandwich structure to concurrently extract high-frequency local sensor transients and long-range temporal dependencies over 10-step telemetry buffers. To eliminate the computational overhead of standard feature attribution methods, we formulate a PyTorch-based Saliency Gradient mapping mechanism that computes normalized explainability scores in 0.78 ms (99.97% faster than Kernel SHAP), satisfying sub-second edge deployment constraints. Predictions exceeding calibrated anomaly thresholds ($\tau > 0.85$) dynamically trigger autonomous host- and network-level active mitigation scripts (via Wazuh-integrated `iptables` filtering and process termination) while concurrently logging structured audit trails to **NIST SP 800-53 Rev. 5** and **ISO/IEC 27001:2022** control frameworks. Extensive empirical validation across all 13 distinct sub-datasets of the ToN_IoT benchmark suite demonstrates state-of-the-art efficacy: 99.70% binary anomaly detection accuracy ($F_1 = 0.9983$) on network flows, and an average 98.37% forensic multi-class accuracy across IoT sensor and host OS domains. Rigorous non-parametric hypothesis testing confirms the statistical superiority of Conformer-Sentinel over competitive GRU, LSTM, and 1D-CNN baselines (Wilcoxon $W = 0.0, p < 10^{-4}$; Friedman $\chi^2_F = 39.00, p < 10^{-7}$; Cohen's $d = 16.09$), while live system profiling validates an end-to-end Mean Time to Respond of $MTTR = 21.5 \text{ ms} \ll 1000 \text{ ms}$, establishing a robust blueprint for autonomous, self-defending industrial cyber-physical infrastructures.

Keywords: Industrial Internet of Things (IIoT), Google Conformer, Extended Detection and Response (XDR), Explainable AI (XAI), Active Mitigation, NIST SP 800-53, ToN_IoT Dataset.

Key Research Contributions

1. **Cross-Layer Heterogeneous Evaluation:** Comprehensive validation across all 13 datasets in the ToN_IoT suite spanning network telemetry, physical IoT sensors (GPS, Modbus, Weather), and host OS kernels (Linux process/disk/memory, Windows 7/10).
2. **Sub-Millisecond Edge Explainability:** A first-order gradient saliency mapping yielding 0.78 ms feature attributions, resolving the latency-explainability trade-off for real-time edge security.
3. **Autonomous Closed-Loop Mitigation:** Real-time Wazuh-integrated active response achieving MTTR = 21.5 ms with zero human intervention.
4. **Automated GRC Governance:** Seamless translation of deep learning anomaly vectors into auditable NIST SP 800-53 and ISO 27001 regulatory compliance matrices.